Quantum Mechanics AY 2025-26, Sem II

Unit–3 :

Concept of Quantum Tunnelling and its applications

Introduction to Quantum Tunnelling

Quantum tunnelling is a purely **quantum mechanical phenomenon**. It has no explanation within the framework of classical mechanics. According to classical physics, a particle cannot cross a region where its energy is less than the potential energy of that region. However, quantum mechanics allows such a process to occur with a certain probability.

The importance of quantum tunnelling is reflected in its many practical applications and scientific discoveries. In 1954, the Nobel Prize in Physics was awarded to **Cockcroft and Walton** for their work in nuclear physics, where tunnelling was used to initiate nuclear reactions that were classically impossible. In 1973, another Nobel Prize was awarded for the development of **tunnel diodes**, electronic devices that operate based on the tunnelling effect. Later, in 1986, **Binnig and Rohrer** received the Nobel Prize for inventing the **Scanning Tunnelling Microscope (STM)**, which made it possible to observe surfaces at the atomic scale using quantum tunnelling.

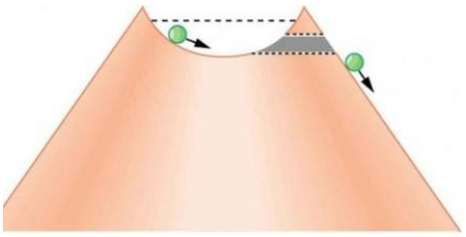


Figure 1: The marble in this semi-circular bowl at the top of a volcano has enough kinetic energy to get to the altitude of the dashed line, but not enough to get over the rim, so that it is trapped forever. If it could find a tunnel through the barrier, it would escape, roll downhill, and gain kinetic energy.

Classical Viewpoint

In classical mechanics, particles are treated as point objects that move along definite and well-defined paths. A particle can overcome an obstacle only if it has enough energy to cross it. If the energy of the particle is less than the height of the barrier, the particle is completely reflected and cannot enter that region.

Classical mechanics does not include any concept of wave behavior or probability. Therefore, there is no mechanism in classical physics that allows a particle to cross a barrier without sufficient energy. As a result, tunneling is strictly forbidden in classical mechanics.

Quantum Mechanical Viewpoint

In quantum mechanics, particles such as electrons, photons, and nucleons do not behave like tiny solid balls. Instead, they exhibit both particle-like and wave-like nature. Each particle is described by a wavefunction that spreads out in space, meaning the particle does not have a sharply defined position.

When a quantum particle encounters a potential barrier, its wave does not stop abruptly at the boundary. Instead, the wave extends slightly into the barrier, even when the particle does not have enough energy to cross it. This leads to a small but non-zero probability of finding the particle inside the barrier.

If the barrier is sufficiently thin, the wave can extend through it and appear on the other side with reduced strength. Consequently, the particle may be detected beyond the barrier. This phenomenon is known as **quantum tunnelling**.

Physical Interpretation of Tunnelling

The fundamental reason for tunnelling lies in the **wave nature of matter**. A useful analogy comes from optics. When a light wave reaches a boundary between two media, part of the wave is reflected and part is transmitted. Even when light falls on an opaque medium, it penetrates a short distance into it.

Similarly, in quantum mechanics, potential energy plays a role analogous to refractive index in optics. When the potential energy changes suddenly, the quantum wave is partly reflected and partly transmitted. This wave behaviour allows particles to enter regions that are forbidden according to classical physics.

Another important aspect is the probabilistic nature of quantum mechanics. Instead of predicting exact paths, quantum theory predicts probabilities of finding particles at different positions. This probabilistic interpretation makes tunnelling possible.

Absence of Classical Proof

Classical mechanics is deterministic in nature and is based on exact trajectories and definite energies. It does not include wave behaviour or probability. Since quantum tunnelling arises entirely due to wave properties and probabilistic

interpretation, classical mechanics cannot explain or predict this phenomenon. Hence, there is no classical proof or explanation of quantum tunneling.

Key Features of Quantum Tunnelling

Quantum mechanics predicts two remarkable effects that are completely absent in classical physics:

- A particle can be reflected even when it has sufficient energy.
- A particle can cross a narrow potential barrier even when its energy is less than the barrier height.

Both of these predictions have been verified experimentally.

Conclusion

Quantum tunnelling is a fundamental consequence of the wave-like and probabilistic nature of particles. It clearly demonstrates the limitations of classical mechanics and highlights the necessity of quantum theory for explaining microscopic phenomena. The successful experimental verification and technological applications of tunnelling confirm its central role in modern physics.

Particle Flux, Reflection and Transmission Coefficients

Particle Flux

Particle flux means the number of particles crossing a given point per unit time. It tells us how fast particles are flowing in a particular direction.

Suppose there are N particles present in one-unit length along the x-direction. If each particle is moving with speed u, then in a small time interval Δt, all the particles present in a length $u\Delta t$ will cross a fixed point.

Therefore, the number of particles crossing the point in time Δt is

$$N \times u\Delta t$$

Hence, the particle flux F, defined as the number of particles crossing per unit time, is given by

$$F = \frac{Nu\Delta t}{\Delta t} = Nu$$

Example: If cars are moving on a road and 10 cars are present in every meter of road, and each car moves at a speed of 5 m/s, then the number of cars crossing a fixed point per second is similar to particle flux.

Wavefunction and Probability

In quantum mechanics, a particle is described by a wavefunction $\psi(x)$. The square of its magnitude, $|\psi(x)|^2$, gives the probability of finding the particle at position x.

The wavefunction can represent:

- a single particle, or
- a large number of identical particles behaving in the same way.

Thus, $|\psi(x)|^2$ can also be interpreted as the **average number of particles per unit length**.

Plane Wave and Moving Particles

Consider the wavefunction $\psi(x) = Ae^{ikx}$

This wavefunction represents a group of particles moving along the positive x-direction with the same momentum.

In this case:

- The exact position of any one particle is not known.
- This uncertainty is allowed and required by the Heisenberg uncertainty principle.
- The average number of particles per unit length is given by $|A|^2$.

Here, the wave number k determines the momentum of the particles.

Particle Speed

The momentum of each particle is proportional to the wave number k. The speed of the particle is obtained from its momentum divided by its mass.

Since k is positive, the speed is also positive, which means the particles are moving in the positive x-direction.

Example: This is similar to water waves moving in one direction in a canal. While the exact water molecule cannot be tracked, the overall flow of water can be clearly measured.

Reflection and Transmission of Particles

When a stream of particles encounters a potential barrier:

- Some particles are reflected back.
- Some particles pass through the barrier.

The **reflection coefficient** represents the fraction of particles that are reflected, while the **transmission coefficient** represents the fraction that passes through the barrier.

These coefficients are calculated by comparing the particle flux of reflected or transmitted particles with the incident particle flux.

Physical Meaning

- Particle flux measures how many particles flow per second.
- Reflection coefficient tells how many particles bounce back.
- Transmission coefficient tells how many particles successfully cross the barrier.

Together, they help us understand how particles behave when they interact with potential barriers in quantum mechanics.

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Particle Flux, Reflection and Transmission Coefficients

Particle Flux for a Plane Wave

Consider a group of quantum particles represented by a plane wave wavefunction

$$\psi(x) = Ae^{ikx}.$$

Such a wavefunction represents many identical particles moving in the same direction with the same momentum. The quantity $|A|^2$ represents the **average number of particles per unit length**. Since this value is constant, the distribution of particles is uniform in space.

The particle flux F is defined as the number of particles crossing a fixed point per unit time. For the above wavefunction, the particle flux is given by

$$F = |A|^2 \frac{\hbar k}{m}$$

This expression shows that particle flux depends on:

• the number of particles present per unit length,

• the speed of the particles.

Physical Meaning: If particles are densely packed (large $|A|^2$), more particles will cross a given point per second. Similarly, if the particles move faster, the particle flux increases.

Example: This is similar to people walking in a corridor. If more people are present per meter and all walk at the same speed, the number of people crossing a door per second will be large.

Important Note on Applicability of the Flux Formula

The above expression for particle flux is valid **only when the wavefunction is a momentum eigenfunction**. This means that:

- all particles have the same momentum,
- the momentum does not vary with position,
- the average number of particles per unit length remains constant.

If the wavefunction represents a superposition of different momenta, then the particle density changes with position, and the simple flux expression cannot be directly applied.

Reflection and Transmission at a Potential Step

When a beam of particles encounters a potential step or barrier, not all particles behave in the same way. Some particles are reflected back, while others cross the potential step and continue moving forward.

To quantitatively describe this behaviour, two important coefficients are defined.

Reflection Coefficient

The **reflection coefficient** R is defined as the ratio of the particle flux of the reflected particles to the particle flux of the incident particles:

$$R = \frac{\text{flux of reflected particles}}{\text{flux of incident particles}}$$

Physical Meaning: The reflection coefficient tells us what fraction of incoming particles bounce back from the potential step.

Example: If 100 particles per second approach a barrier and 30 particles per second are reflected, then the reflection coefficient is $R = 0.3$.

Transmission Coefficient

The **transmission coefficient** T is defined as the ratio of the particle flux of the transmitted particles to the particle flux of the incident particles:

$$T = \frac{\text{flux of transmitted particles}}{\text{flux of incident particles}}$$

Physical Meaning: The transmission coefficient represents the fraction of particles that successfully cross the potential step or barrier.

Example: If out of 100 incident particles per second, 70 particles per second cross the barrier, then the transmission coefficient is $T = 0.7$.

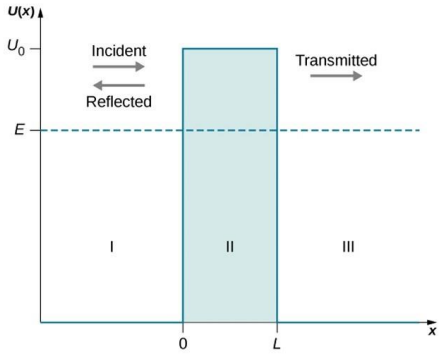


Figure 2: Finite potential barrier

Conservation of Particles

In quantum mechanics, particles are neither created nor destroyed during reflection and transmission processes. Therefore, the total number of particles remains conserved.

As a result, the reflection and transmission coefficients must satisfy the condition:

$$R + T = 1$$

Interpretation: This condition simply means that every incident particle is either reflected or transmitted. There is no third possibility.

Reflection and Transmission at a Potential Barrier

Consider a steady beam of electrons (or other quantum particles) moving along the positive x-direction with energy E. The particles encounter a potential barrier of finite width L and height U_0. The potential energy is zero outside the barrier and constant inside the barrier. **Case 1: Infinite Barrier Height**

If the height of the potential barrier is infinite, the quantum particle cannot enter the barrier. The wavefunction becomes zero at the boundary, and the particle is completely reflected. This situation is similar to classical mechanics, where a particle bounces back from an infinitely high wall.

Case 2: Infinite Barrier Width and Finite Height

If the barrier has finite height but infinite width, the quantum wave can penetrate slightly into the barrier. However, the wave gradually decreases inside the barrier and eventually vanishes. In this case, no particle emerges on the other side, and transmission does not occur.

Case 3: Finite Barrier Height and Finite Width

When both the height and width of the barrier are finite, the incident quantum wave partially penetrates the barrier and decreases gradually inside it. If the barrier is thin enough, part of the wave reaches the far side and emerges as a transmitted wave. This phenomenon is known as **quantum tunneling**.

The amount of tunneling depends on the height and width of the barrier and on the energy of the incident particle. A higher or wider barrier reduces tunneling, while higher particle energy increases it.

The probability of reflection and transmission of particles is obtained by solving the time-independent Schr¨odinger equation with appropriate boundary conditions at the barrier edges. This analysis explains how quantum particles can cross barriers that are forbidden in classical physics.

Reflection and Transmission at a Finite Potential Barrier

Consider a particle of mass m and energy E incident along the positive xdirection on a one-dimensional finite potential barrier of height U_0 and width L, defined as

$$V(x) = \begin{cases} 0, & x < 0 \quad \text{(Region I)} \\ U_0, & 0 \le x \le L \quad \text{(Region II)} \\ 0, & x > L \quad \text{(Region III)} \end{cases}$$

The behavior of the particle is obtained by solving the time-independent Schr¨odinger equation in each region.

Time-Independent Schro¨dinger Equation

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

Solution in Different Regions

Region I (*x*< 0)

Here $V = 0$, so the Schr¨odinger equation becomes

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \quad \text{where } k = \sqrt{\frac{2mE}{\hbar^2}}$$

The general solution is

$$\psi_I = Ae^{ikx} + Be^{-ikx}$$

where:

- Ae^{ikx} represents the incident wave,
- Be^{-ikx} represents the reflected wave.

Region II ($0 \leq x \leq L$**)**

Case 1: $E < U_0$ **(Tunneling Case)**

Here the Schr¨odinger equation becomes

$$\frac{d^2\psi}{dx^2} - \kappa^2\psi = 0 \quad \text{where } \kappa = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}}$$

The solution is

$$\psi II = Ce^{\kappa x} + De^{-\kappa x}$$

This solution represents an exponentially decaying wave inside the barrier.

Region III (*x*>*L*)

Again $V = 0$, so

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0$$

The solution is

$$\psi III = Fe^{ikx}$$

This represents the transmitted wave moving in the positive x-direction.

Boundary Conditions

The wavefunction and its first derivative must be continuous at the boundaries $x = 0$ and $x = L$:

- $\psi_I(0) = \psi_{II}(0)$
- $\psi_I'(0) = \psi_{II}'(0)$
- $\psi_{II}(L) = \psi_{III}(L)$
- $\psi_{II}'(L) = \psi_{III}'(L)$

Using these conditions, the constants are related and the transmission and reflection coefficients are obtained.

Transmission and Reflection Coefficients

The particle flux is proportional to the square of the wave amplitude multiplied by the particle velocity.

Reflection Coefficient

$$R = \frac{\text{Reflected flux}}{\text{Incident flux}} = \frac{|B|^2}{|A|^2}$$

Transmission Coefficient

$$T = \frac{\text{Transmitted flux}}{\text{Incident flux}} = \frac{|F|^2}{|A|^2}$$

Transmission Coefficient for Tunneling ($E < U_0$)

After applying boundary conditions, the transmission coefficient is found to be

$$T = \frac{1}{1 + \frac{U_0^2}{4E(U_0 - E)} \sinh^2(\kappa L)}$$

Reflection Coefficient

Since particle number is conserved,

$$R + T = 1 \quad \Rightarrow \quad R = 1 - T$$

Discussion of Tunnelling

- Even when $E < U_0$, the transmission coefficient is non-zero.
- Tunnelling probability decreases as barrier width L increases.
- Tunnelling probability decreases as barrier height U_0 increases.
- Tunnelling probability increases with particle energy E.

Conclusion

Quantum tunnelling arises due to the wave nature of particles. By solving the Schr¨odinger equation for a finite potential barrier, we find that particles have a finite probability of crossing a classically forbidden region. This result has no classical analogue and is a fundamental prediction of quantum mechanics.

Interpretation: The transmission probability is extremely small but nonzero, confirming that quantum tunneling is possible even when $E < U_0$, though highly unlikely for a thick and high barrier.

Realistic Examples of Quantum Tunnelling

Quantum tunnelling is a purely quantum mechanical phenomenon in which a particle has a finite probability of crossing a potential barrier even when its total energy E is less than the barrier height U_0. This occurs because the particle is described by a wavefunction ψ, which penetrates and decays exponentially inside the barrier instead of becoming zero abruptly as predicted by classical physics.

For a one-dimensional barrier, the wavefunction inside the barrier ($E < U_0$) is given by

$$\psi(x) = Ae^{-\kappa x}, \quad \kappa = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}}$$

This leads to a non-zero transmission probability.

Alpha Decay

Alpha decay is a natural example of quantum tunnelling in nuclear physics. An alpha particle (helium nucleus) is confined inside the atomic nucleus by a strong nuclear attractive force and a repulsive Coulomb potential barrier.

Protons and neutrons are tightly held inside the nucleus by strong nuclear force, so energy is needed to remove them. This situation is like a marble inside a bowl. The marble can move inside the bowl, but if it does not have enough energy, it cannot cross the edge and will remain trapped. In the same way, particles inside the nucleus do not have enough energy to escape. However, in alpha decay, an alpha particle (two protons and two neutrons) still comes out of the nucleus. This happens because of a quantum effect called tunneling, explained by George Gamow. In quantum mechanics, particles behave like waves, so there is a small chance that the alpha particle can pass through the energy barrier and escape, even without enough energy.

Inside the barrier, the wave function does not become zero but decreases exponentially, and we do not observe the particle inside the barrier. The probability of finding a particle is related to the square of its wave function, and

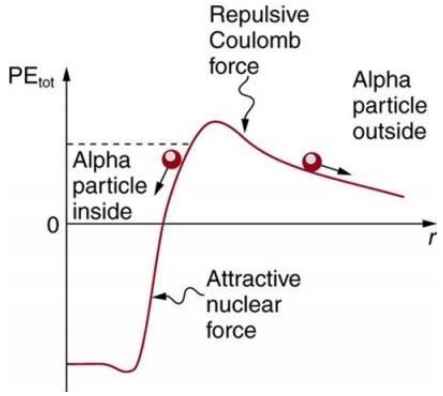


Figure 3: Nucleons within an atomic nucleus are bound or trapped by the attractive nuclear force, as shown in this simplified potential energy curve. An alpha particle outside the range of the nuclear force feels the repulsive Coulomb force. The alpha particle inside the nucleus does not have enough kinetic energy to get over the rim, yet it does manage to get out by quantum mechanical tunnelling.

so there is a small probability of finding the particle outside the barrier, which implies that the particle can tunnel through the barrier. This process is called barrier penetration or quantum mechanical tunnelling.

- Inside the nucleus, the alpha particle has energy E less than the Coulomb barrier height U_0.

- Classically, escape from the nucleus is impossible since $E < U_0$. • Quantum mechanically, the alpha particle wavefunction penetrates the barrier:

$$\psi(r) \propto e^{-\kappa r}$$

- The tunnelling probability is approximately:

$$T \propto e-2\kappa L$$

  where L is the width of the nuclear barrier.

- This tunnelling probability determines the half-life of radioactive nuclei.

Scanning Tunnelling Microscope (STM)

The scanning tunnelling microscope is a practical application of quantum tunnelling used to observe surfaces at the atomic scale.

- A sharp metallic tip is brought within a distance of about 1nm from a conducting surface.

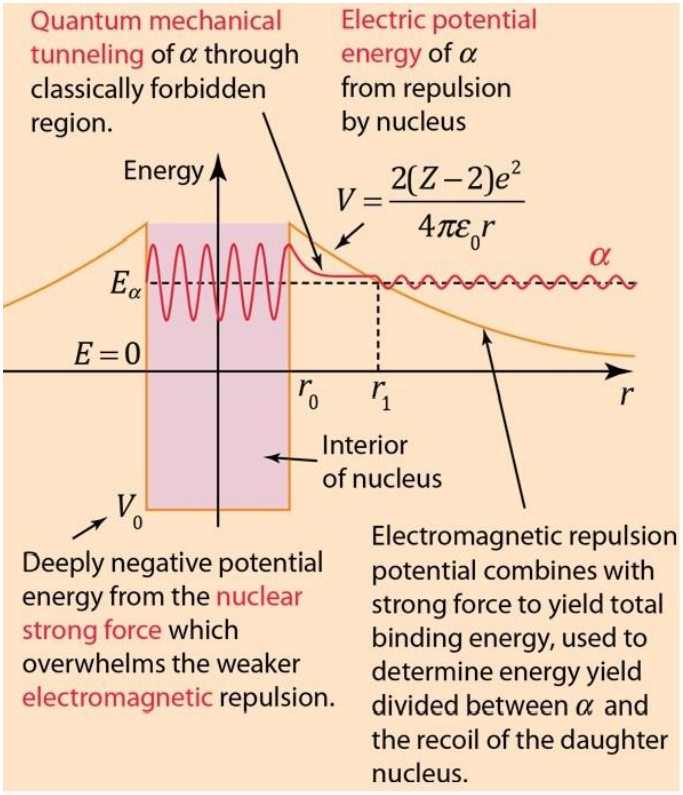


Figure 4: Modeling Polonium-212 Alpha Half-life

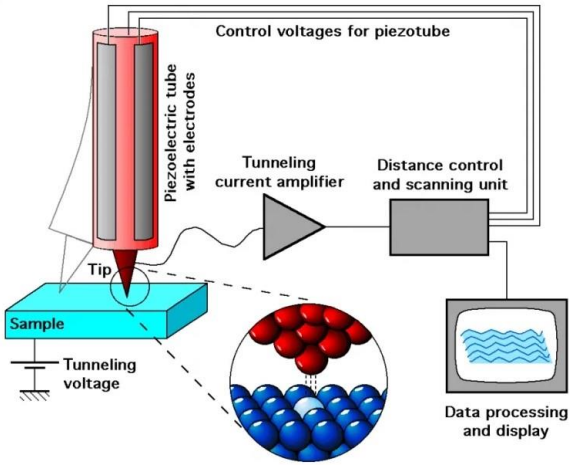


Figure 5: Schematic structure of a scanning tunneling microscope

- A small bias voltage *V* is applied between the tip and the surface.

- Electrons tunnel through the vacuum barrier separating the tip and the surface.
- The tunnelling current I depends exponentially on the tip-surface separation d:

$$I \propto e^{-2\kappa d}$$

- Here,

$$\kappa = \sqrt{\frac{2m\phi}{\hbar^2}}$$

  where ϕ is the effective work function.
- A change of 0.1nm in distance produces a large change in current, allowing atomic resolution.

Key Physical Insights

- The exponential nature of tunnelling explains why barrier width strongly affects tunnelling probability.
- Higher particle energy increases tunnelling probability.
- Heavier particles have lower tunnelling probability due to larger mass m.
- Tunnelling has no classical explanation and confirms the wave nature of matter.

Conclusion

Quantum tunnelling explains important natural processes such as alpha decay and enables advanced technologies like the scanning tunnelling microscope. The mathematical treatment using the Schr¨odinger equation shows that tunnelling probability decreases exponentially with barrier width and height, highlighting the fundamentally non-classical behaviour of microscopic particles.

Simple Harmonic Oscillator (1D) : Quantum Explanation

A simple harmonic oscillator is a system in which a particle oscillates back and forth about an equilibrium position due to a restoring force. In quantum mechanics, this model is used to describe vibrations of atoms and molecules.

Potential Energy

In one dimension, the potential energy of a harmonic oscillator is

$$U = \frac{1}{2}kx^2$$

where

- k is the force constant which is related to the frequency of oscillation ν and mass m of the particle by the relation,

$$\nu = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

- x is the displacement from equilibrium.

Schro¨dinger Equation

Substituting $V(x)$ into the Schr¨odinger equation:

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{1}{2}kx^2\right)\psi = 0 \qquad (1)$$

Physical Meaning and Applications

- The first term represents kinetic energy.
- The second term represents potential energy.
- Solving this equation gives quantized energy levels.
- Vibrational spectra of diatomic molecules
- Phonons in solids (quantized lattice vibrations)
- Quantum field theory (as the free field Hamiltonian)
- Quantum optics (electromagnetic field quantization)

Eq. (1) can be simplified by introducing the dimensionless quantities

$$y = \left(\frac{1}{\hbar}\sqrt{km}\right)^{1/2} x \text{ and } \alpha = \frac{2E}{h\nu} \qquad (2)$$

The purpose of introducing these quantities in Eq. (1) is to change the units in x and E to dimensionless units.
Schrödinger's equation in terms of y and α is given by

$$\frac{d^2\psi}{dy^2} + (\alpha - y^2)\psi = 0 \qquad (3)$$

The acceptable solutions to this equation are limited by the condition that $\psi \to 0 \; as \; y \to \infty$. This condition is fulfilled when

α = 2n + 1, where n = 0, 1, 2, 3, ... (4)

Solution and Energy Levels

Substituting the value of α from Eq. (2) in Eq. (4), the energy levels of a harmonic oscillator with frequency of oscillation ν is given by

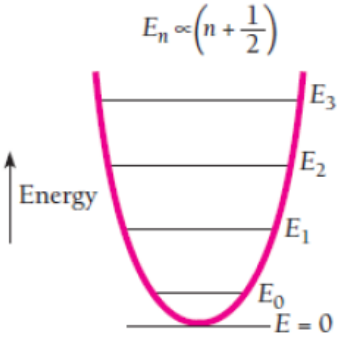
$$E_n = \left(n + \frac{1}{2}\right) h\nu, \quad n = 0, 1, 2, 3, \ldots. \qquad (5)$$

where *n* is the quantum number.

The ground state energy

$$E_0 = \frac{1}{2} h\nu$$

represents the zero-point energy, a purely quantum mechanical phenomenon.



- Energy levels are equally spaced.
- The minimum energy is not zero.

- The lowest value of energy of a harmonic oscillator is called zero-point energy because a harmonic oscillator in equilibrium with its surroundings will approach this value and not E = 0 as temperature approaches 0K.

Wave Functions (Eigenfunctions)

The general formula for the nth normalized wave function of a harmonic oscillator in terms of the Hermite polynomial $H_n(y)$is given by

$$\psi_n = \left(\frac{2mv}{\hbar}\right)^{1/4} (2^n n!)^{-1/2} H_n(y) e^{-y^2/2} \qquad (6)$$

The lowest-order Hermite polynomials are given below:

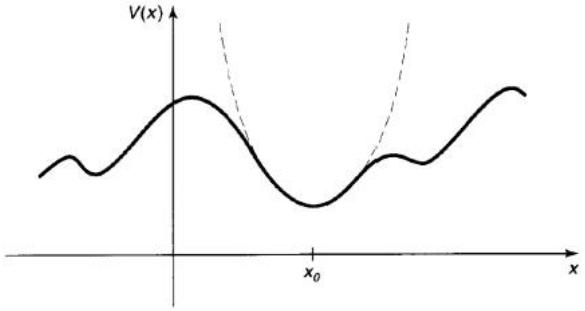
$$H_0(y) = 1;\ H_1(y) = 2y;\ H_2(y) = 4y^2 - 2;\ H_3(y) = 8y^3 - 12y$$

Physical Meaning

- The particle cannot be completely at rest due to quantum uncertainty.
- Even in the lowest energy state, the particle has some motion.
- The oscillator vibrates about the equilibrium position $x = 0$.

Significance of harmonic oscillator in quantum mechanics

The significance of harmonic oscillator in quantum mechanics is that any particle that is in its position of a stable equilibrium will execute simple harmonic motion (SHM) if displaced by a small amount. For example, simple pendulum or atom of hydrogen in hydrogen chloride (HCl).
Intermolecular potential energy vs. intermolecular distance for a diatomic molecular distance is shown in the figure alongside.



Parabolic approximation (dashed curve) to an arbitrary potential, in the neighborhood of a local minimum.

Difference from Classical Oscillator

- Classical oscillator can have zero energy.
- Quantum oscillator always has minimum energy.
- Classical energy is continuous; quantum energy is discrete.

Conclusion

The one-dimensional quantum harmonic oscillator shows that energy is quantized and that a particle always possesses some minimum energy. This simple model is essential for understanding vibrations in atoms and molecules.

Quantum Theory of Hydrogen Atom (Simple Mathematics)

The hydrogen atom consists of one proton at the nucleus and one electron moving around it. Quantum mechanics explains the structure and energy levels of the hydrogen atom more accurately than classical theories.

Schro¨dinger Equation

The motion of the electron is described by the Schr¨odinger equation. Solving this equation gives allowed energy levels and wavefunctions.

Time-Independent Schro¨dinger Equation

The Schrödinger's equation for the electron in three dimensions i.e., hydrogen atom is

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{2m}{\hbar^2}(E-U)\psi=0 \qquad (1)$$

Here, U represents the electric potential energy between the proton and electron separated by a distance r and is given by

$$U=-\frac{e^2}{4\pi\varepsilon_0 r} \qquad (2)$$

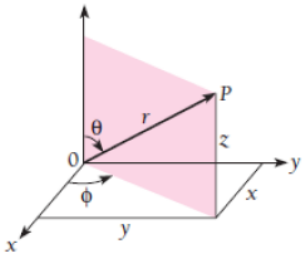
Substituting Eq. (2) in Eq. (1), we get

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{2m}{\hbar^2}\left(E+\frac{e^2}{4\pi\varepsilon_0 r}\right)\psi=0 \qquad (3)$$

Physical Meaning

- The equation describes the motion of the electron around the nucleus.
- Solutions give allowed energy levels and orbitals.
- The electron is described by probability, not fixed paths.

It is difficult to solve Eq. (3) in Cartesian co-ordinates. However, if the Cartesian co-ordinates (x, y, z) are transformed to spherical polar co-ordinates, (r, θ, ϕ). then Eq. (3) can be easily solved.



Here, r is the length of the radius vector from the origin O to the point P.
θ represents the angle between the radius vector and +z axis. It is termed as the zenith angle.
ϕ represents the angle between the projection of the radius vector in the xy plane and +x axis. It is termed as the azimuthal angle.
The Cartesian co-ordinates are related to spherical co-ordinates by the following relations.

$$r=\sqrt{x^2+y^2+z^2}$$

$$x=r\sin\theta\cos\phi$$

$$y=r\sin\theta\sin\phi$$

$$z=r\cos\theta$$

The Schrödinger's equation in spherical polar co-ordinates is given by

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}+\frac{2m}{\hbar^2}\left(E+\frac{e^2}{4\pi\varepsilon_0 r}\right)\psi=0 \qquad (4)$$

The above equation can be separated easily into three independent equations, each involving only one variable. After carrying out appropriate substitutions and separating the variables, the following three total differential equations are obtained

Azimuthal wave equation

$$\frac{d^2\Phi}{d\phi^2} + m_l^2\Phi = 0 \qquad (5)$$

Polar wave equation

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \left[l(l+1) - \frac{m_l^2}{sin^2\theta}\right]\Theta = 0 \qquad (6)$$

Radial wave equation

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{2m}{\hbar^2}\left(\frac{e^2}{4\pi\varepsilon_o r} + E\right) - \frac{l(l+1)}{r^2}\right]R = 0 \qquad (7)$$

The solution of the azimuthal wave equation (5) is given by

$$\Phi(\phi) = Ae^{im_l\phi} \qquad (8)$$

The condition that the wave function Φ must have a single value at a given point in space. Now, since ϕ and $\phi + 2\pi$ represent a single point in space $Ae^{im_l\phi} = e^{im_l(\phi+2\pi)}$

This is possible only if m_l is 0 or a positive or negative integer.

i.e., $m_l = 0, \pm1, \pm2, \pm3, \ldots\ldots\ldots$

The constant m_l is known as the magnetic quantum number.

The polar wave equation (6) has a complex solution in terms of associated Legendre functions provided that the constant $l$in Eq. (6) is an integer equal to or greater than the absolute value of the magnetic quantum number i.e., $|m_l|$. This implies

$m_l = 0, \pm1, \pm2, \pm3, \ldots\ldots\ldots, \pm l.$

The constant l is termed as the orbital quantum number.

The radial wave equation (7) has a complex solution in terms of associated Laguerre polynomials under the following conditions.

(i) The energy E in eq. (7) have one of the negative values E_n(signifying that the electron is bound to the atom) given by

$$E_n = -\frac{me^4}{32\pi^2\varepsilon_0^2\hbar^2}\left(\frac{1}{n^2}\right) = \frac{E_1}{n^2} n = 1, 2, 3, \ldots. \quad (9)$$

(ii) The constant n in Eq. (9), known as the principal quantum number must be equal to or greater than $l + 1$.This condition means that

$$l = 0, 1, 2, 3, \ldots\ldots\ldots, (n-1)$$

Hence, the three quantum numbers needed to describe the three-dimensional motion of the electron in the hydrogen atom are n, l and m_l and they must satisfy the following conditions.

(a) Principal quantum number, $n \geq 1; n = 1, 2, 3, \ldots.$
(b) Orbital quantum number, $l \leq (n-1); l = 0, 1, 2, 3, \ldots\ldots\ldots\ldots\ldots, (n-1)$
(c) Magnetic quantum number, $|m_l| \leq l;\ m_l = 0, \pm1, \pm2, \pm3, \ldots\ldots\ldots, \pm l$

The wave function for the ground state of hydrogen atom is

$$\psi_1 = \left(\frac{1}{\sqrt{\pi}a_0^{3/2}}\right)e^{-r/a_0} \quad (10)$$

where a_0 is the radius of the innermost orbit of the hydrogen atom also termed as first Bohr radius.

The probability P(r)dr of finding an electron at any radial distance within a spherically symmetric shell of thickness dr and volume $4\pi r^2 dr$ is given by

$$P\,(r)dr = |\psi_n|^2 4\pi r^2 dr$$

The probability of finding the electron in the ground state of the hydrogen atom is given by

$$P\,(r)dr = |\psi_1|^2 4\pi r^2 dr = \psi_1^* \psi_1 4\pi r^2 dr = \left[\frac{1}{\pi a_0^3}e^{-2r/a_0}\right]4\pi r^2 dr$$

Hydrogen-like Ions

For ions with nuclear charge *Ze* (He^+, Li^{2+}, etc.), replace:

- $a_0 \rightarrow a_0/Z$ in radial functions
- $e^2 \rightarrow Ze^2$ in potential
- $E_n \rightarrow -\frac{Z^2}{n^2} \times 13.6\text{eV}$

Physical Interpretation

- The wave functions describe **orbitals**—probability distributions for finding the electron
- Nodes: Radial nodes = $n - \ell - 1$, angular nodes = ℓ
- The 1s ground state has no nodes (maximum at nucleus)
- Higher ℓ states have zero probability at nucleus
- The exponential decay ensures bound states

Mathematical Notes

The solution involves:

1. Separation of variables: $\psi(r,\theta,\phi) = R(r)\Theta(\theta)\Phi(\phi)$
2. Solving the radial equation using associated Laguerre polynomials
3. Solving the angular equation using spherical harmonics
4. Quantization emerges from boundary conditions:
    - $\Phi(\phi)$ single-valued: m integer
    - $\Theta(\theta)$ finite: ℓ integer, $|m| \leq \ell$
    - $R(r)$ normalizable: n integer, $n > \ell$

These wave functions form a complete orthonormal set:

$$\int_0^\infty r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi \, \psi^*_{n\ell m} \psi_{n'\ell'm'} = \delta_{nn'}\delta_{\ell\ell'}\delta_{mm'}$$

Atomic Orbitals

- The electron does not move in fixed circular orbits.
- It is described by a wavefunction ψ.
- $|\psi|^2$ gives the probability of finding the electron.
- Regions of high probability are called atomic orbitals.

Quantum Numbers

The state of the electron is described by quantum numbers:

- Principal quantum number n ($n = 1,2,3,...$),
- Orbital quantum number l ($0 \leq l \leq n - 1$),
- Magnetic quantum number m ($-l \leq m \leq l$).

Emission and Absorption of Radiation

When an electron jumps from a higher energy level n_2 to a lower level n_1, a photon is emitted:

$$hv = En2 - En1$$

- This explains the line spectrum of hydrogen.
- Different transitions produce different spectral lines.

Energy Levels

The allowed energy levels are

$$E_n = -\frac{13.6}{n^2} \text{ eV} \qquad (n = 1,2,3,...)$$

- Negative energy means the electron is bound to the nucleus.
- Transitions between levels explain hydrogen spectrum.

Conclusion

Quantum theory successfully explains the stability, energy levels, and emission spectrum of the hydrogen atom. The use of simple energy formulas and probability concepts makes it a fundamental model in quantum mechanics.

The Schr¨odinger equation provides a simple mathematical framework to understand quantum systems like the harmonic oscillator and the hydrogen atom. By choosing appropriate potential energy functions, it explains energy quantization and atomic structure.

The hydrogen atom wave functions provide the quantum mechanical description of atomic structure, explaining atomic spectra, chemical bonding, and serving as the foundation for understanding multi-electron atoms through the Aufbau principle and periodic table structure.

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